



COVENANT UNIVERSITY
CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc. Degree Examination

COLLEGE: College of Science and Technology

DEPARTMENT: Computer and Information Sciences

SESSION: 2020/2021

COURSE CODE: CSC213

COURSE TITLE: Structured Programming

INSTRUCTION: Answer any **FOUR(4)** Questions

SEMESTER: ALPHA

CREDIT UNIT: 3

TIME: 3 Hours

QUESTION ONE (17.5 MARKS)

- a. What is the major difference between a *Pseudocode* and actual *Python code*? (1 Mark)
- b. What will be the output of the Following?

```
months = {'Jan':1, 'Feb':2, 'Mar':3}
print (months[2])
```

i.

(1 Mark)

```
print('_\nX\tO\tX\nO\tX\tO\nX\tO\tX\n_')
```

ii.

(1 Mark)

- c. Correct the errors and write the output of the following piece of code: (4.5 Marks)

```
def squar(a)
    c=a**2
    return c

number = 4
x = []

sqr = square(number)
sqr2 = square(sqr)
x[0] = square(sqr2)

print ("The Square of" number, "is", sqr)
print ("The Square of" sqr, "is", sqr2)
print ("The Square of" sqr2, "is", x[0])
```

- d. The Covenant Technologies sells computer accessories and have declared a five-day sale. Each day, starting on Monday, the price of all items will drop 10% of the previous day's price. For example, if the original price of a product is N2000.00, the sale price on Monday would be N1800.00 (10% less than the original price); on Tuesday the sale price would be N1620.00 (10% less than Monday) etc. Write a python programme that will compute the price of any item for each of the five days, given the original price. Use a **For Loop**, a **value-returning Function** named **Discount** to compute the discount and a Nested Lists **SalesDays** to hold the days and computed discounts. The Final output **Must** be formatted as follows:

QUESTION TWO (17.5 MARKS)

- a. What will **list1** and **list2** contain after the following statements are executed?

(1 Mark)

```
list1 = ['F', 'U'] * 2
list2 = [100]
list2 += list1
```

- b. Write a python program that displays the number of times a user supplied character appears in a given text file named "file1.txt" using the Count function. Your program should inform the user if the character entered does not appear in the file. The program should allow the user count as many times as he/she wants. NB: Assume that the file is in your current working directory and count function syntax is `str.count(x)`. (7 Marks)
- c. The local driver's license office has asked you to create an application that grades the written portion of the driver's license exam. The exam has 20 multiple-choice questions. Here are the correct answers:

1. A	6. B	11. A	16. C
2. C	7. C	12. D	17. B
3. A	8. A	13. C	18. B
4. A	9. C	14. A	19. D
5. D	10. B	15. D	20. A

Your program should store these correct answers in a **List**. The program should accept the student's answers for each of the 20 questions from the keyboard and store the answers in another list. The program should display a message indicating whether the student passed or failed the exam. (A student must correctly answer 15 of the 20 questions to pass the exam.) It should then display the **total number** of correctly answered questions, the **total number** of incorrectly answered questions, and a **list** showing the question numbers of the incorrectly answered questions. (9.5 Marks)

QUESTION THREE (17.5 MARKS)

- a. Mention how Python Tuples are immutable (1 Mark)
- b. What is an Infinite loop? Show a relevant code example (2 Mark)
- c. Given the List PCdata below

```
PCdata=[
    ["Acer", "4GB RAM", "500GB HDD", "Corei7"],
    ["HP", "8GB RAM", "1TB HDD", ["Corei7", "Corei5", "Corei3"], "17Inches Display"],
    ["Dell", "4GB RAM", "750GB HDD", "Corei5"],
    ["Lenovo", "300GB HDD", "12Inches Display"],
    ["Sony", "2GB RAM", "Mini Series"],
    ["Surface", "5TB HDD", ["Pro", "RT"]]
]
```

What would be the output of the following print statements?

- i. `print (PCdata[1])`
- ii. `print (PCdata [1][3])`
- iii. `print (len(PCdata))`
- iv. `print (len(PCdata[3])+len(PCdata[5][2]))`
- v. `print (PCdata[3]+PCdata[5][2])`
- vi. `print (PCdata[0].append('Corei9'))`

(6.5 Marks)

- d. Using at least 4 Functions and the Random Module, write a program that generates printable addition, subtraction, multiplication, and division tests. The tests should consist of 12 questions each of addition, subtraction, multiplication, and division questions (in the order) in the following format: **Question 1(Addition): num1 + num2 = Answer**. The question number goes from 1 to 12, and num1 and num2 are randomly generated integer numbers between 1 and 10. The program should simply display the 12 questions and their solutions. **NB: It should not prompt the user for any input.** (8 Marks)

QUESTION FOUR (17.5 MARKS)

- a. Given `mySet = set('one two')`; what will be the output of `print(mySet)`? (1 Mark)
- b. What values will the following code display? (1 Mark)

```
myDict = {1:[0, 1], 2:[2, 3], 3:[4, 5]}
for k in myDict:
    print(k)
```

- c. Assume each of the variables `set1` and `set2` references a set. Write code that creates another set containing only the elements that are found in both `set1` and `set2`, and assigns the resulting set to the variable `set3`. (2 Marks)
- d. Write a python program to create a dictionary called "grocery_list" which stores the name and quantity of things you want your parents to send you this weekend. Write the dictionary to an already existing file in your current working directory named "myshoppinglist.txt". (4.5 Marks)
- e. Suppose a file named **ChapelServiceAttendance.txt** contains the headers FName, LName, Matric No and four chapel attendance statuses for each student in the month of November. "1" signifies presence while "0" means the student was absent. The data is delimited by a tab:

FName	LName	Matric No.	3Nov	10Nov	17Nov	24Nov
Nnamdi	Chukwu	12CF1122	1	1	0	1
Aderopo	Ude	12CF2233	0	0	1	1
Shina	Kudi	11CA4431	1	1	1	1
Helen	Mike	12CD8877	0	1	1	1
Kyle	Tunde	12CV7766	0	0	0	1

Write a python programme to read the content of the file and display the attendance percentage for all students on the screen: (Hint: compute Attendance percentage as (No. of times present/4) * 100)
e.g. Nnamdi Chukwu 12CF1122 75%. (9 Marks)

QUESTION FIVE (17.5 MARKS)

- a. Mention **One(1)** major difference between C Language and Python (1 Mark)
- b. Given the List of some books of the Bible:

```
BibleBooks=['Matthew', 'Mark', 'Luke', 'John', 'Psalms', 'Romans']
```

Write the Python Code and Use List methods to achieve the following:

- Remove **Psalms** from the list.
- Get the index of **Luke**.
- Insert **Acts** in the list after **John**.
- Sort the list.
- Reverse the order of items in the List
- Extend the list by adding **1 Corinthians** and **2 Corinthians** as a Nested List to the List
- What will be the eventual output of `print (BibleBooks)`? (7 Marks)

- c. You have a group of friends coming to visit for your high school reunion, and you want to take them out to eat at a local restaurant. You aren't sure if any of them have dietary restrictions, but your restaurant choices are as follows:

Restaurant	Vegetarian	Vegan	Gluten-Free
Sheraton Hotels	Yes	No	No
Eko Hotels	No	Yes	Yes
Protea Hotels	No	No	Yes
Goshen	Yes	No	Yes
Apples	No	Yes	No
Cora	No	Yes	Yes
Dome	No	No	No
Hooters	Yes	No	No
Jollibee	Yes	Yes	Yes
Five Guys	Yes	Yes	No
CU Cafeteria	Yes	Yes	Yes

Using Lists, write a program that asks whether any members of your party are vegetarian, vegan, or gluten-free, to which then displays only the restaurants to which you may take the group. Here is an example of the program's output:

Is anyone in your party a vegetarian? Yes

Is anyone in your party a vegan? Yes

Is anyone in your party gluten-free? Yes

Here are your restaurant choices: Jollibee, CU Cafeteria (9.5 Marks)

QUESTION SIX (17.5 MARKS)

- How many iterations would occur in total if a loop that repeats 5 times is nested within a loop that repeats 4 times? (1 Mark)
- What is a Boolean function? Show a relevant Code example (2 Marks)
- Write a programme to determine whether the length of a string entered by the user is in the range of 10 to 15 and the First letter in the string is Capitalised. If the length of the string is outside this range and the first letter is not capitalised it should display the relevant error message, i.e. "Invalid Word: First Letter not Titled" or "Invalid Word: No Long Enough". Otherwise, it should display "Valid Word". (4 Marks)
- Write a python program to perform the following: (10.5 Marks)
 - Receive the following information from the user about their cousins: First name, Last name, Age, State of Origin and Gender.
 - Write the details to a tab-delimited file called "cousins.txt".
 - Write another program to read from the "cousins.txt" file and display the details of the oldest friend.